4-25-14

Physical Chemistry

Kinetics

1. (25 pts) Isopropenyl allyl ether in vapor state isomerizes to allyl acetone according to a first-order rate equation. The following equation gives the influence of temperature on the rate constant (in s^{-1}).

$$k = 5.4 \times 10^{11} e^{-123000/RT}$$

where the activation energy is expressed in J mol^{-1}. At 150 °C, how long will it take to build up a partial pressure of 0.395 bar of allyl acetone, starting with 1 bar of isopropenyl allyl ether?

2. (25 pts) If a first-order reaction has an activation energy of 104600 J mol^{-1} and, the in the equation $k = Ae^{-E_a/RT}$, A has a value of 5×10^{13} s^{-1}, at what temperature will the reaction have a half-life of 1 min?

3. (50 pts) The photochemical oxidation of phosgene, sensitized by chlorine, has been studied. The overall reaction is

$$2COCl_2 + O_2 \longrightarrow 2CO_2 + 2Cl_2$$

and the rate expression that gives the effect of the several variables is

$$\frac{d(CO_2)}{dt} = \frac{kI_0(COCl_2)}{1 + k'(Cl_2)/(O_2)}$$

where I_0 is the intensity of the light. The elementary steps were proposed to involve the following reactions:

$$COCl_2 + h\nu \xrightarrow{k_1} COCl + Cl$$

$$COCl + O_2 \xrightarrow{k_2} CO_2 + ClO$$

$$COCl_2 + ClO \xrightarrow{k_3} CO_2 + Cl_2 + Cl$$

$$COCl + Cl_2 \xrightarrow{k_4} COCl_2 + Cl$$

$$Cl + Cl \xrightarrow{k_5} Cl_2$$

Derive the expression of k and k'.